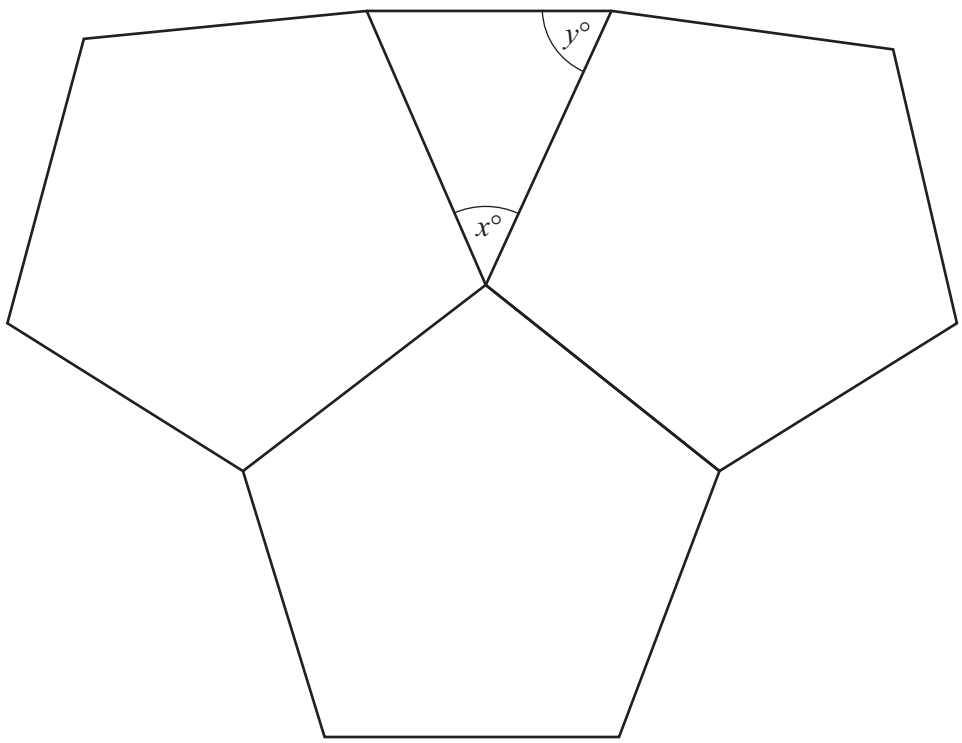
**1** The diagram shows three congruent regular pentagons and a triangle.




NOT TO SCALE

**(a)** Work out the value of $x$.
Give a geometrical reason for your answer.

$x$ = ..... because .....

..... [2]

**(b)** Work out the value of $y$.
Give a geometrical reason for your answer.

$y$ = ..... because .....

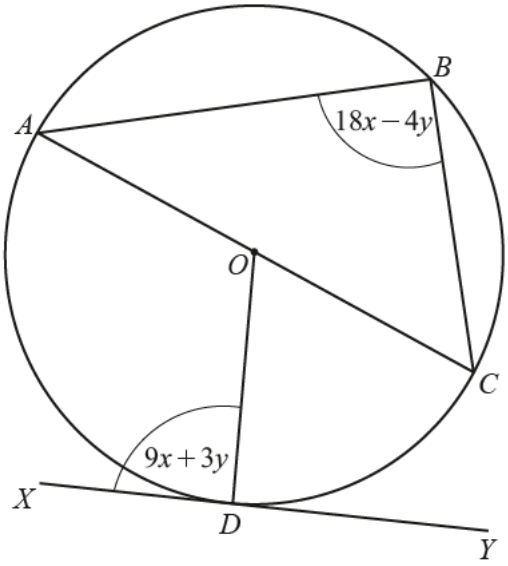
..... [3]

**(c)** Find the ratio $x : y$.
Give your answer in its simplest form.

..... : ..... [1]

[Total: 6]

**2** In this question, all angles are in degrees.




NOT TO SCALE

$A$, $B$, $C$ and $D$ lie on a circle, centre $O$, diameter $AC$.
$XY$ is a tangent to the circle at $D$.

**(a)** Use the information in the diagram to complete these two simultaneous equations.

$9x + 3y =$ .....

$18x - 4y =$ ..... [2]

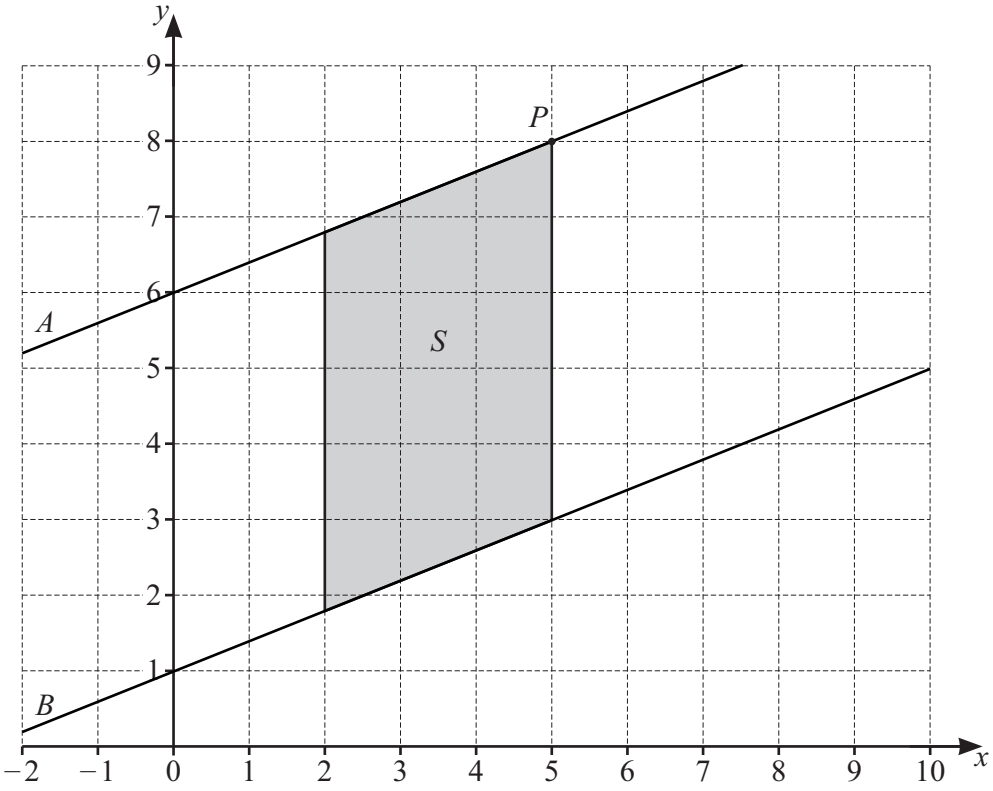
**(b)** Solve your simultaneous equations.
You must show all your working.

$x =$ .....

$y =$ ..... [3]

[Total: 5]

**3** The diagram shows a point $P$, a shape $S$ and lines $A$ and $B$ on a 1 $cm^2$ grid.




**(a)** Line $A$ is parallel to line $B$.

Explain what parallel means.

..... [1]

**(b)** Write down the coordinates of point $P$.

( ..... , ..... ) [1]

**(c)** **(i)** Write down the mathematical name for shape $S$.

..... [1]

**(ii)** Work out the area of shape $S$.

..... $cm^2$ [1]

**(d)** **(i)** Find the gradient of line $A$.

..... [1]

**(ii)** Write down the equation of line $A$.

..... [2]

[Total: 7]

**4** A box, in the shape of a cuboid, has volume $357\,\text{cm}^3$.
It has a length of 8.5 cm and a width of 6 cm.

Calculate the height of the box.

..... cm [2]

[Total: 2]

**5** An empty fuel tank is filled using a cylindrical pipe with diameter 8 cm.
Fuel flows along this pipe at a rate of 2 metres per second.
It takes 24 minutes to fill the tank.

Calculate the capacity of the tank.
Give your answer in litres.

..... litres [4]

[Total: 4]

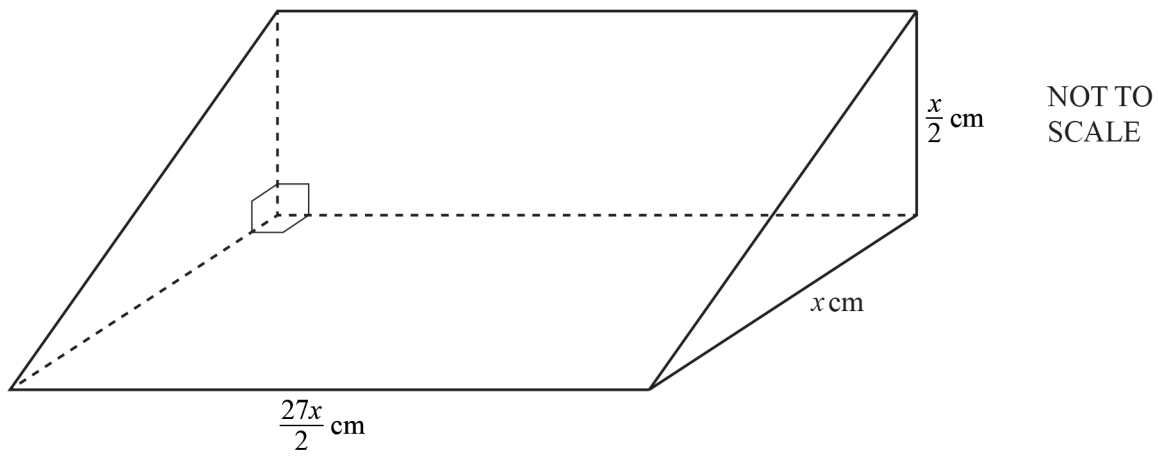
**6** A cuboid has a square base.
The volume of this cuboid is $867\,cm^3$ and its height is $12\,cm$.

Calculate the length of one side of the square base.

.....cm [3]

[Total: 3]

**7**




A prism with a right-angled triangle as its cross-section has a volume of $1000\,\text{cm}^3$.

Calculate the value of $x$.

$x =$ ..... [4]

[Total: 4]

**8** The volume of a solid metal sphere is $24\,430\,\text{cm}^3$.

**(a)** Calculate the radius of the sphere.

[The volume, $V$, of a sphere with radius $r$ is $V = \frac{4}{3}\pi r^3$.]

..... cm [3]

**(b)** The metal sphere is placed in an empty tank.
The tank is a cylinder with radius 50 cm, standing on its circular base.
Water is poured into the tank to a depth of 60 cm.

Calculate the number of litres of water needed.

..... litres [3]

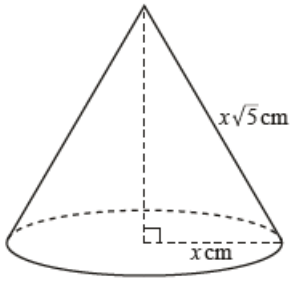
[Total: 6]

**9** A tank is a cuboid measuring 1.8 m by 1.5 m by 1.2 m.
Water flows from a pipe into this empty tank at a rate of 200 $cm^3$ per second.

Find the time it takes to fill the tank.
Give your answer in hours and minutes.

..... hours ..... minutes [4]

[Total: 4]




NOT TO SCALE

**10** A cone with radius $x$ cm and slant height $x\sqrt{5}$ cm has a volume of 1000 $cm^3$.

Calculate the value of $x$.

[The volume, $V$, of a cone with radius $r$ and height $h$ is $V = \frac{1}{3}\pi r^2 h$.]

$x =$ ..... [4]

[Total: 4]